



Name:	
Student ID:	



Final Examination

Phys332: Electromagnetic Theory II

2026/06/10

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

- (1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;
- (2) I take *full responsibility* for any ambiguity in my selections in “multiple choice questions”;
- (3) incorrect selections will receive $-1/7$ of the question's points;
- (4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;
- (5) no partial credit is awarded for the explanations provided in the answer boxes;
- (6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded negative points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;
- (7) any page which does not contain *both my name and student id* may not be graded;
- (8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: _____

Question:	1	2	3	4	5	6	7	Total
Points:	21	21	$10\frac{1}{2}$	$10\frac{1}{2}$	21	21	0	105

Name:	
Student ID:	



All dimensionfull quantities are to be understood in SI units.

(For instance, $\epsilon_0 \simeq 10^{-11}$ and $|\mathbf{B}| = 5$ stand for $\epsilon_0 \simeq 10^{-11} \text{ A}^2 \text{ kg}^{-1} \text{ m}^{-3} \text{ s}^4$ and $|\mathbf{B}| = 5 \text{ A}^{-1} \text{ kg}^1 \text{ s}^{-2}$.)

Question: 1: Electromagnetic field of time dependent current (21 points)
(MIDTERM THREE QUESTION RELOADED)

Please feel free to use the identity $\frac{d}{dx} \log(-b - 2x + 2\sqrt{a^2 + bx + x^2}) = -(a^2 + bx + x^2)^{-1/2}$.

Consider an electrically-neutral infinite straight wire in an otherwise empty space. Assume that there is no current on the wire until $t = 0$; and, at time $t = 0$, a current of constant magnitude 4π gets abruptly turned on. Mathematically, $I(t) = 4\pi\theta(t)$ for the Heaviside-theta function θ . Set up a coordinate system such that any point on the wire has $x = y = 0$ and the current flows in $(-\hat{z})$ direction.

(a) ($10^{1/2}$ points) In Lorenz gauge, we can derive the vector potential as

$$\mathbf{A}(\mathbf{r}, t) = \mu_0 \hat{z} \int_{\alpha}^{\beta} \frac{dz}{\sqrt{z^2 + r^2 + 2Cz\mathbf{r} \cdot \hat{z}}} \quad \text{for} \quad \begin{cases} \alpha = B\mathbf{r} \cdot \hat{z} - A\sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{z})^2} \\ \beta = \mathbf{r} \cdot \hat{z} + A\sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{z})^2} \end{cases} \quad (1)$$

What is the tuple (A, B, C) ?

- ☐ $(-1, -1, -1)$
☐ $(-1, -1, 1)$
☒ $(-1, 1, -1)$
☐ $(-1, 1, 1)$
- ☐ $(1, -1, -1)$
☐ $(1, -1, 1)$
☐ $(1, 1, -1)$
☐ $(1, 1, 1)$

(b) ($10^{1/2}$ points) For an arbitrary unit vector $\hat{n}$, we can derive $\mathbf{A}(ct\hat{n}, t) = \mu_0 \hat{z} \log \left(\frac{1 + A|\hat{n} \cdot \hat{z}|^C}{1 + B|\hat{n} \cdot \hat{z}|^C} \right)$ in the Lorenz gauge. What is the tuple (A, B, C) ?

- ☐ $(-1, -1, 1)$
☐ $(-1, -1, 2)$
☒ $(-1, 1, 1)$
☐ $(-1, 1, 2)$
- ☐ $(1, -1, 1)$
☐ $(1, -1, 2)$
☐ $(1, 1, 1)$
☐ $(1, 1, 2)$

Solution 1.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), we start by deriving the vector potential. In Lorenz gauge,

$$\mathbf{A}(\mathbf{r}, t) = -\frac{\mu_0}{4\pi} \hat{z} \int_{-\infty}^{\infty} \frac{I\left(t - \frac{|\mathbf{r} - z\hat{z}|}{c}\right)}{|\mathbf{r} - z\hat{z}|} dz \quad (2)$$

where we oriented our coordinate system such that the wire is in the $-z$ -direction. We can now insert the formula for I :

$$\mathbf{A}(\mathbf{r}, t) = -\mu_0 \hat{z} \int_{-\infty}^{\infty} \frac{\theta\left(t - \frac{|\mathbf{r} - z\hat{z}|}{c}\right)}{|\mathbf{r} - z\hat{z}|} dz \quad (3)$$



Name:	
Student ID:	

If $ct < |\mathbf{r} - z\hat{\mathbf{z}}|$, the argument of Heaviside-theta becomes negative, hence the integrand becomes zero, hence we are restricted to the integration range for $ct > |\mathbf{r} - z\hat{\mathbf{z}}|$ which means

$$(ct)^2 > r^2 + z^2 - 2z\mathbf{r} \cdot \hat{\mathbf{z}} \Rightarrow (ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2 > (z - \mathbf{r} \cdot \hat{\mathbf{z}})^2 \quad (4)$$

hence

$$\mathbf{r} \cdot \hat{\mathbf{z}} + \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2} > z > \mathbf{r} \cdot \hat{\mathbf{z}} - \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2} \quad (5)$$

meaning

$$\mathbf{A}(\mathbf{r}, t) = -\mu_0 \hat{\mathbf{z}} \int_{\mathbf{r} \cdot \hat{\mathbf{z}} - \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2}}^{\mathbf{r} \cdot \hat{\mathbf{z}} + \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2}} \frac{dz}{\sqrt{z^2 + r^2 - 2z\mathbf{r} \cdot \hat{\mathbf{z}}}} \quad (6)$$

which can be rewritten as

$$\mathbf{A}(\mathbf{r}, t) = \mu_0 \hat{\mathbf{z}} \int_{\mathbf{r} \cdot \hat{\mathbf{z}} + \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2}}^{\mathbf{r} \cdot \hat{\mathbf{z}} - \sqrt{(ct)^2 - r^2 + (\mathbf{r} \cdot \hat{\mathbf{z}})^2}} \frac{dz}{\sqrt{z^2 + r^2 - 2z\mathbf{r} \cdot \hat{\mathbf{z}}}} \quad (7)$$

For part (b), we have

$$\mathbf{A}(ct\hat{\mathbf{n}}, t) = \begin{cases} -\mu_0 \hat{\mathbf{z}} \int_0^{2ct\hat{\mathbf{n}} \cdot \hat{\mathbf{z}}} \frac{dz}{\sqrt{z^2 + (ct)^2 - z(2ct\hat{\mathbf{n}} \cdot \hat{\mathbf{z}})}} & \text{for } \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} > 0 \\ -\mu_0 \hat{\mathbf{z}} \int_{2ct\hat{\mathbf{n}} \cdot \hat{\mathbf{z}}}^0 \frac{dz}{\sqrt{z^2 + (ct)^2 - z(2ct\hat{\mathbf{n}} \cdot \hat{\mathbf{z}})}} & \text{for } \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} < 0 \end{cases} \quad (8)$$

both of which can be combined into a single equation (by changing dummy variable $z \rightarrow -z$ and integration orientation in the second case):

$$\mathbf{A}(ct\hat{\mathbf{n}}, t) = -\mu_0 \hat{\mathbf{z}} \int_0^{2ct|\hat{\mathbf{n}} \cdot \hat{\mathbf{z}}|} \frac{dz}{\sqrt{z^2 + (ct)^2 - z(2ct|\hat{\mathbf{n}} \cdot \hat{\mathbf{z}}|)}} \quad (9)$$

We are given the relation

$$\frac{1}{\sqrt{a^2 + bx + x^2}} = -\frac{d}{dx} \log(-b - 2x + 2\sqrt{a^2 + bx + x^2}) \quad (10)$$

which we can integrate and use the fundamental theorem of calculus to get

$$\int_{\alpha}^{\beta} \frac{dx}{\sqrt{x^2 + a^2 + bx}} = \log \left(\frac{-b - 2\alpha + 2\sqrt{a^2 + b\alpha + \alpha^2}}{-b - 2\beta + 2\sqrt{a^2 + b\beta + \beta^2}} \right) \quad (11)$$

Name:	
Student ID:	



specifically

$$\int_0^{-b} \frac{dx}{\sqrt{x^2 + a^2 + bx}} = \log \left(\frac{-b + 2a}{b + 2a} \right) \quad (12)$$

Comparing this with (9), we conclude

$$A(ct\hat{n}, t) = -\mu_0 \hat{z} \log \left(\frac{2ct|\hat{n} \cdot \hat{z}| + 2ct}{-2ct|\hat{n} \cdot \hat{z}| + 2ct} \right) \quad (13)$$

which simplifies to

$$A(ct\hat{n}, t) = \mu_0 \hat{z} \log \left(\frac{1 - |\hat{n} \cdot \hat{z}|}{1 + |\hat{n} \cdot \hat{z}|} \right) \quad (14)$$

Apart from the μ_0 , which is only there for the particular choice of the unit system, the result is independent of c or t ! Although it is harder to understand why with our 3-dimensional approach, this is natural from a 4-dimensional perspective: the result is purely geometric as it shows the so-called lightcone structure.

Question: 2: Analysis of radiation in the temporal gauge (21 points)

(MIDTERM THREE QUESTION RELOADED)

Define radiation as the part of the electromagnetic fields that can carry energy to the spatial infinity.

We then define radiation power at t as $P(t) := \lim_{R \rightarrow \infty} \oint_{S_R^2} \mathbf{S}(\mathbf{r}, t + |\mathbf{r}|/c) \cdot d\mathbf{A}$ for S_R^2 denoting a sphere of

radius R where $\mathbf{S}(\mathbf{r}, t)$ is the Poynting vector at the position $\mathbf{r}$ and time t . In the so-called *temporal gauge*, we choose $V = 0$ under which an explicit derivation leads to the conclusion that the most dominant piece in $\mathbf{S}$ as r goes to infinity is

$$\mathbf{S}(\mathbf{r}, t_r) \supset \left[\hat{r} |\mathbf{A}'(\mathbf{r}, t_r)|^2 + \alpha (\mathbf{A}'(\mathbf{r}, t_r) \cdot \hat{r}) \mathbf{A}'(\mathbf{r}, t_r) \right] / c \quad (15)$$

for some constant α and for the shorthand notation $\mathbf{A}'(\mathbf{r}, t_r) := \frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r}$, and that only this part can actually contribute to the radiation. Which statement below is then completely correct?

- ☐ We can now use $\mathbf{A}(\mathbf{r}, t) \propto \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{|\mathbf{r} - \mathbf{r}'|} d^3 \mathbf{r}'$ to compute the total radiation power.
- ☐ An arbitrary time dependence of $\mathbf{A}$ would suffice for nonzero radiation.
- ☒ **Only the transverse component of $\mathbf{A}$ can contribute to the radiation.**
- ☐ Radiation computed this way is the extreme of all possible radiations subject to the gauge choice.
- ☐ If we used Coulomb gauge instead of temporal gauge, we would find precisely the same functional form of $\mathbf{A}$.
- ☐ There is no residual gauge transformation left, as we completely fixed $V = 0$.



Name:	
Student ID:	

- ☐ Poynting vector being quadratic in A is a special property of the temporal gauge, and is absent in other (for instance radiation) gauges.
- ☐ This whole question is incorrect, because our very definition of radiation power was done in Lorenz gauge, and then we tried to analyze it in a completely different gauge!

Solution 2.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

We are told that we should work in a gauge where $V = 0$: this is of course possible, i.e. a suitable λ can always be found to make $V = 0$ via

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla \lambda, \quad V \rightarrow V - \frac{\partial \lambda}{\partial t} \quad (16)$$

In this gauge, we have

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A} \quad (17)$$

hence

$$\mathbf{S}(\mathbf{r}, t) = -\frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} \times (\nabla \times \mathbf{A}(\mathbf{r}, t)) \quad (18)$$

For $t_r := t - \frac{|\mathbf{r}|}{c}$ denoting the retarded time, we have

$$\mathbf{S}(\mathbf{r}, t_r) = -\frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r} \times (\nabla \times \mathbf{A}(\mathbf{r}, t_r)) \quad (19)$$

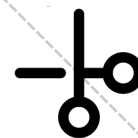
Now, in general, the curl would act both on $\mathbf{r}$ and t_r ; however, the first piece dies out faster at infinity hence can not contribute to the radiation. To understand the second piece, we note that

$$\frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial x^i} \supset \frac{\partial \mathbf{A}(\mathbf{r}, t_r)}{\partial t_r} \frac{\partial t_r}{\partial x^i} \Rightarrow \partial_i A_j(\mathbf{r}, t_r) \supset -\frac{1}{c} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \frac{x_i}{r} \quad (20)$$

hence

$$\begin{aligned} S_i(\mathbf{r}, t_r) &= -\epsilon_{ijk} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \epsilon_{klm} \partial_\ell A_m(\mathbf{r}, t_r) \\ &\supset \frac{1}{c} \epsilon_{ijk} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \epsilon_{klm} \frac{\partial A_m(\mathbf{r}, t_r)}{\partial t_r} \frac{x_\ell}{r} \\ &\supset \frac{1}{c} (\epsilon_{kij} \epsilon_{klm}) \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \frac{\partial A_m(\mathbf{r}, t_r)}{\partial t_r} \frac{x_\ell}{r} \\ &\supset \frac{1}{c} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \frac{x_i}{r} - \frac{1}{c} \frac{\partial A_j(\mathbf{r}, t_r)}{\partial t_r} \frac{\partial A_i(\mathbf{r}, t_r)}{\partial t_r} \frac{x_j}{r} \end{aligned} \quad (21)$$

Name:	
Student ID:	



yielding

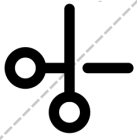
$$S(\mathbf{r}, t_r) \supset \frac{\hat{r} |A'(\mathbf{r}, t_r)|^2 - (A'(\mathbf{r}, t_r) \cdot \hat{r}) A'(\mathbf{r}, t_r)}{c} \quad (22)$$

for the shorthand notation $A'(\mathbf{r}, t_r) := \frac{\partial A(\mathbf{r}, t_r)}{\partial t_r}$. Let's now look at the given statements:

1. We can now use $A(\mathbf{r}, t) \propto \int \frac{J(\mathbf{r}', t_r)}{|\mathbf{r} - \mathbf{r}'|} d^3 \mathbf{r}'$ to compute the total radiation power.
This formula is derived from wave equation for a decoupled A which was constructed thanks to the Lorenz gauge: it is not applicable in this different gauge!
2. An arbitrary time dependence of A would suffice for nonzero radiation.
Not necessarily; in reality, the terms due to a single time derivative of the sources will cancel each other in their contribution to the radiation, we need the second order time derivatives (such as accelerated sources).
3. Only the transverse component of A can contribute to the radiation.
Look at the formula we have derived, (22): we see that part of the A that is parallel to $\hat{r}$ cancels between the two terms: longitudinal part of A does not contribute to the radiation!
4. Radiation computed this way is the extreme of all possible radiations subject to the gauge choice.
Radiation is integral of Poynting vector which only depends on E , B , and μ_0 : all these three are gauge invariant quantities, hence this statement does not make sense!
5. If we used Coulomb gauge instead of temporal gauge, we would find precisely the same functional form of A .
Let's go over gauge conditions: temporal gauge ($V = 0$), Coulomb gauge ($\nabla \cdot A = 0$), radiation gauge ($\nabla \cdot A = \frac{\partial V}{\partial t} = 0$), and Lorenz gauge ($\nabla \cdot A + \epsilon_0 \mu_0 \frac{\partial V}{\partial t} = 0$). Obviously, they are not independent; for instance, if we are in radiation gauge, then Lorenz gauge condition is already satisfied. However, temporal gauge itself does not imply Coulomb gauge, so there is no reason to make that claim!
6. There is no residual gauge transformation left, as we completely fixed $V = 0$.
We know from (27) that $A \rightarrow A + \nabla \lambda$ with $V \rightarrow V - \frac{\partial \lambda}{\partial t}$ for any λ gives us a gauge transformation. If we choose λ to be time-invariant, then this gauge transformation preserves temporal gauge condition, hence it is a residual gauge transformation.
7. Poynting vector being quadratic in A is a special property of temporal gauge, and is absent in other (for instance radiation) gauges.
This is simply incorrect and can be quickly checked by dimensional analysis!
8. This whole question is incorrect, because our very definition of radiation power was done in Lorenz gauge, and then we tried to analyze it in a completely different gauge!
Radiation power depends only on gauge-invariant quantities, hence its definition can not be taken specific for a particular gauge choice!

Question: 3: Radiation Gauge in Electromagnetic Theory ($10\frac{1}{2}$ points)
(AND CONCEPT OF RESIDUAL GAUGE TRANSFORMATION)

In class, we have learnt of the *radiation gauge*, where a system in this gauge obeys both the Coulomb



Name:	
Student ID:	

and Lorenz gauge constraints. Although not all systems can be brought to the radiation gauge, assume that we deal with an electromagnetic system for which this is possible. If such a system is in radiation gauge and if we apply the gauge transformation

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\lambda, \quad V \rightarrow V - \frac{\partial\lambda}{\partial t} \quad \text{for} \quad \lambda(x, y, z, t) = (X^T A X) t \quad \text{for} \quad X \doteq \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (23)$$

which option below for the matrix A ensures that the system remains in the radiation gauge?

Hint: As we have discussed in class, this is actually a question about residual gauge transformation.

- ☐ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

☒ $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

☐ $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

Solution 3.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

We are told that the radiation gauge is combination of Lorenz gauge, i.e.

$$\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} = 0 \quad (24)$$

and the Coulomb gauge, i.e.

$$\nabla \cdot \mathbf{A} = 0 \quad (25)$$

hence the radiation gauge is basically the constraint

$$\frac{\partial V}{\partial t} = 0, \quad \nabla \cdot \mathbf{A} = 0 \quad (26)$$

As the most general gauge transformation is

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\lambda, \quad V \rightarrow V - \frac{\partial\lambda}{\partial t} \quad (27)$$

for an arbitrary scalar field λ , we have

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\lambda, \quad V \rightarrow V - \frac{\partial\lambda}{\partial t} \quad \text{for scalar field } \lambda \text{ such that } \nabla^2 \lambda = \frac{\partial^2 \lambda}{\partial t^2} = 0 \quad (28)$$

Name:	
Student ID:	



We are given λ in the form

$$\lambda(x, y, z, t) = t \left(X^T A X \right) \quad \text{for} \quad X \doteq \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (29)$$

which trivially satisfies $\frac{\partial^2 \lambda}{\partial t^2} = 0$. Note that all non-diagonal pieces of the matrix A comes with cross terms (e.g. xy) whose laplacian is zero, therefore the laplacian only acts on the diagonal terms for which it gives $\nabla^2 x^2 = \nabla^2 y^2 = \nabla^2 z^2 = 2$, hence

$$\nabla^2 \lambda = 0 \quad \Rightarrow \quad \text{Tr}(A) = 0 \quad (30)$$

which is satisfied by a single option.

Question: 4: Basics of Special Relativity (10½ points)
(AT THE LEVEL WE UNDERSTOOD IN THIS COURSE)

Please choose the correct option below.

- ☐ Proper time is a *Lorentz scalar*, hence it is invariant under coordinate transformations between any two frames.

- ☐ We can understand why radiation is only sourced by the accelerated charges via the relativistic argument that four acceleration $\frac{d^2 X^\mu}{d\tau^2}$ is always orthogonal to four velocity $\frac{dX^\mu}{d\tau}$, i.e. $\frac{d^2 X^\mu}{d\tau^2} \frac{dX_\mu}{d\tau} = 0$.

- ☐ For a Gallilean transformation $x_i \rightarrow x'_i$ between two inertial frames, the partial derivative $\frac{\partial x'_i}{\partial x_j}$ is always equal to a rotation matrix.

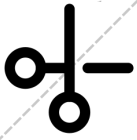
- ☐ In special relativity, any frame in which light is measured to have the speed c is an inertial frame!

- ☒ **There is no way to continuously change a real velocity from subliminal range ($v < c$) to superliminal range ($v > c$).**

- ☐ Special relativity ensures that no object can move faster than light in the vacuum!

- ☐ Gallilean transformations form the unique set of transformations that are compatible with the existence of an absolute time!

- ☐ There always exist an inertial frame in which all electromagnetic fields are purely of electric nature (i.e. no magnetic field).



Name:	
Student ID:	

Solution 4.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

1. Lorentz scalars are invariant under Lorentz transformations, which only relate two inertial frames not any generic two frames.
2. The four acceleration is perpendicular to four velocity for any motion in spacetime; for instance, even for a particle at rest in an inertial frame has a four acceleration perpendicular to four velocity! Hence the argument does not follow.
3. This is true only if both inertial frames have the same handedness: for a transformation from a right handed inertial frame to a left handed inertial frame, we instead have $\frac{\partial x'_i}{\partial x_j} = -R_i^j$.
4. For a frame to be inertial, we also need the light to move rectilinearly!
5. Indeed, unless we work with a complex velocity, the only way to go from $v < c$ to $v > c$ is a disconnected jump (no continuous motion)!
6. All special relativity tells us that anything that goes with the speed of light for one observer would go at that speed for any other observers; in particular, it does not say something can not go faster than light! If we further assume causality, then we can not allow information to progress faster than light; however, individual particles can go faster than light if they do not carry information (this is already used in quantum field theory, as we explicitly discussed in the class)!
7. Existence of absolute time and the assumption that Newton's first law holds in all inertial frames uniquely yield the Galilean transformations; however, if Newton's law was different (or we start with a different assumption), we would get a different set of transformations even if we had absolute time!
8. This is true for any given local point; so for a given point, we can always remove magnetic field! But we can not do it for extended regions, i.e. for a whole inertial frame!

Question: 5: Relativistic Kinematics (21 points)

Lightcone coordinates $x_{\pm}$ and rapidity ϕ are to be defined as $x_{\pm} = x \pm ct$ and $\phi = \frac{1}{2} \log \left(\frac{c+v}{c-v} \right)$ respectively.

Super Proton Synchrotron (SPS) and Large Hadron Collider (LHC) are particle accelerators located at CERN (*Conseil Européen pour la Recherche Nucléaire* - European Organization for Nuclear Research) that can accelerate charged particles, usually protons, to relativistically relevant speeds. ATLAS is a particle detector that can analyze the properties of particles released at the collision of protons accelerated at LHC. Define the following coordinate systems, which are assumed to be inertial for the purposes of this question:

ℒ: The Lab frame, i.e. the one in which the CERN gift shop is at rest. By extension, any buildings or detectors such as ATLAS are at rest in the Lab frame.

Name:	
Student ID:	



- p1: The frame in which *the proton beam one*, some protons in SPS, are at rest.
 p2: The frame in which *the proton beam two*, some protons in LHC, are at rest.
 p3: The frame in which *the proton beam three*, some protons in ATLAS detector, are at rest.

Now, consider the following statements:

- (A) : p3 can always be rotated such that propagation direction of any electromagnetic plane wave at the ATLAS detector can be described by $x_+ = 0$ if the medium of ATLAS satisfies $\{\epsilon, \mu\} = \{\epsilon_0, \mu_0\}$.
 (B) : p3 can always be rotated such that propagation direction of any electromagnetic plane wave at the ATLAS detector can be described by $x_+ = 0$ if the medium of ATLAS satisfies $\{\epsilon, \mu\} \neq \{\epsilon_0, \mu_0\}$.
 (C) : If all proton beams are moving along the x -axis in $\mathcal{L}$ and if we know proton beam one and proton beam two has the speeds $\frac{e^{14} - 1}{e^{14} + 1}c$ and $\frac{e^{10} - 1}{e^{10} + 1}c$ in $\mathcal{L}$ respectively, the rapidity of proton beam one in p2 is 4.
 (D) : If all proton beams are moving along the x -axis in $\mathcal{L}$ and if we know proton beam one and proton beam two has the speeds $\frac{e^{14} - 1}{e^{14} + 1}c$ and $\frac{e^{10} - 1}{e^{10} + 1}c$ in $\mathcal{L}$ respectively, the rapidity of proton beam one in p2 is 2.
 (E) : In 2011, an experiment (called OPERA) claimed that an elementary particle called neutrino (created via a process that starts at SPS) actually moved faster than the speed of light: it later turned out to be an incorrect result of equipment failure, but let's assume that the story was true and that special relativity is still valid. Within these assumptions, for such superluminal neutrinos, we can never arrange the proton beam one such that the neutrinos are at rest in p1.
 (F) : Under the very same conditions of (E), we can actually arrange proton beam two such that the superluminal neutrinos are at rest in p2.
 (a) (**10^{1/2} points**) Which one among the options is the largest set that only contains true statements?

- ☐ $\emptyset$ ☒ $\{A\}$ ☐ $\{B\}$ ☐ $\{C\}$ ☐ $\{A, B\}$ ☐ $\{A, C\}$ ☐ $\{B, C\}$ ☐ $\{A, B, C\}$

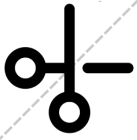
- (b) (**10^{1/2} points**) Which one among the options is the largest set that only contains true statements?

- ☐ $\emptyset$ ☐ $\{D\}$ ☐ $\{E\}$ ☐ $\{F\}$ ☒ $\{D, E\}$ ☐ $\{D, F\}$ ☐ $\{E, F\}$ ☐ $\{D, E, F\}$

Solution 5.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Let's go over the statements:

1. A & B: Since ATLAS detector is at the Lab frame $\mathcal{L}$, we can always rotate p3 such that its minus x -axis matches the direction of photon in $\mathcal{L}$: this is possible because Lorentz transformations map lines to lines (they are linear transformations). Then, in p3, light has $x = -vt$, hence $x_+ = (c - v)t$. It becomes zero only in the vacuum, i.e. when $v = c$.



Name:	
Student ID:	

2. C & D: Given information tells us that

(a) Rapidity of p_1 with respect to $\mathcal{L}$ is $\frac{1}{2} \log \left(\frac{1 + \frac{e^{14}-1}{e^{14}+1}}{1 - \frac{e^{14}-1}{e^{14}+1}} \right) = 7$

(b) Rapidity of p_2 with respect to $\mathcal{L}$ is $\frac{1}{2} \log \left(\frac{1 + \frac{e^{10}-1}{e^{10}+1}}{1 - \frac{e^{10}-1}{e^{10}+1}} \right) = 5$

In class, we have learned that rapidity satisfies the additivity property under Lorentz transformations, just as velocity satisfies this property in Gallilean transformations, i.e. $\phi_{AC} = \phi_{AB} - \phi_{CB}$. Therefore, the rapidity of p_1 with respect to p_2 is $7 - 5 = 2$.

3. E & F: As we have extensively discussed in class, continuous Lorentz transformations can not convert a speed $v > c$ to a speed $v < c$. Therefore there is no rest frame for a superluminal particle, as that would mean one observer can see $v > c$ for the same object whereas another observer can see $v = 0$.

Question: 6: Cascaded Lorentz Transformations (21 points)

(WE EXPLICITLY DISCUSSED THIS IN CLASS AS THE THOMAS ROTATION.)

This question has direct conceptual relevance to atomic physics and the history of relativistic quantum mechanics through the so-called spin-orbit coupling. However, as I can not expect you to know quantum physics, we will consider a fictitious scenario with direct analogy instead.

Consider *Albert, Bob, Charlie, David, and Eva*, each using their own inertial frame to measure the current density in a wire that is at rest with respect to Albert. In Albert's frame, the current is zero for $t < 0$ and a switch gets closed at $t = 0$, and Albert measures a constant current density $\mathbf{J}$ for $t > 0$.

The inertial frames of all these observers are chosen such that they observe the switch getting closed at $t = 0$ (in their own time parameter) and such that the switch is momentarily at the origin of their inertial frame at the time of getting closed. We also know that

1. Bob's frame is boosted along y -axis of Albert's frame with rapidity ϕ .
2. Charlie's frame is boosted along x -axis of Bob's frame with rapidity ϕ .
3. David's frame is boosted along x -axis of Albert's frame with rapidity ϕ .
4. Eva's frame is boosted along y -axis of David's frame with rapidity ϕ .

Finally, each boost is chosen to be the standard boost in the sense that it reduces to the identity as $\phi \rightarrow 0$, so that all frames coincide with Albert's frame in that limit.

Answer the questions below while assuming that ϕ is sufficiently small such that the so-called *Baker–Campbell–Hausdorff formula* can be approximated as

$$e^{\phi_A} e^{\phi_B} \simeq e^{\phi_B} e^{\phi_A} e^{\phi^2[A,B]} \quad (31)$$

Hint: In class, we have derived the general form of arbitrary boosts (remember $e^{\phi K_z}$ etc.) and discussed how they do not commute: this is a question about that!

- (a) (**10 $\frac{1}{2}$ points**) What is the absolute value of the y -component of the current density measured by Eva in her coordinate system, if Charlie measures $\mathbf{J} = \hat{y}$ in his coordinate system?

Name:	
Student ID:	



☐ 0
 ☐ 1
 ☒ $\cos(\phi^2)$
 ☐ $\sin(\phi^2)$
 ☐ $\tan(\phi^2)$
 ☐ $\cosh(\phi^2)$
 ☐ $\sinh(\phi^2)$
 ☐ $\tanh(\phi^2)$

(b) ($10^{1/2}$ points) What is the absolute value of the y -component of the current density measured by Eva in her coordinate system, if Charlie measures $\mathbf{J} = \hat{z}$ in his coordinate system?

☒ 0
 ☐ 1
 ☐ $\cos(\phi^2)$
 ☐ $\sin(\phi^2)$
 ☐ $\tan(\phi^2)$
 ☐ $\cosh(\phi^2)$
 ☐ $\sinh(\phi^2)$
 ☐ $\tanh(\phi^2)$

Solution 6.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

We have derived in class that the most general Lorentz transformation takes the form

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} \quad \Lambda \doteq \mathcal{P}^m \mathcal{T}^n \exp \left(\sum_{i \in \{x,y,z\}} K_i \phi_i + R_i \theta_i \right) \quad (32)$$

for $m, n = 0, 1$ where $\mathcal{P}$ and $\mathcal{T}$ denote the parity and time reversal transformations respectively. The special case $m = n = 0$ is the proper orthochronous Lorentz transformations, and other cases lead to transformations between inertial frames that can not be continuously connected, such as between a left handed and a right handed inertial frames. As we are told that all frames coincide with Albert's frame as $\phi \rightarrow 0$ (a continuous deformation), the relevant transformations have to be proper orthochronous.

We also derived and detailed the boost generators K_i and rotation generators R_i ; explicitly,

$$K_x \doteq \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_y \doteq \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_z \doteq \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad (33a)$$

and

$$R_z \doteq \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad R_y \doteq \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad R_x \doteq \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (33b)$$

Note that you need not to remember the conventions, i.e. you will still find the correct result if you define, say, R_y with an overall sign. However, which one is a boost or which one is a rotation about z -axis is not a convention, hence you should know that to solve this question.

With all the information given in the question, we can now write down the connection between the current $\mathbf{J}$ between observers. For that, consider the four-current J^{μ} whose spatial part is $\mathbf{J}$. This four-current will



Name:	
Student ID:	

transform as a four-vector, hence given information means

$$\begin{aligned}
 J_B^\mu &= e^{(K_y)^\mu{}_\nu \phi} J_A^\nu \\
 J_C^\mu &= e^{(K_x)^\mu{}_\nu \phi} J_B^\nu \\
 J_D^\mu &= e^{(K_x)^\mu{}_\nu \phi} J_A^\nu \\
 J_E^\mu &= e^{(K_y)^\mu{}_\nu \phi} J_D^\nu
 \end{aligned} \tag{34}$$

Obviously, we get $J_A^\mu = J_B^\mu = J_C^\mu = J_D^\mu = J_E^\mu$ for $\phi \rightarrow 0$ as all coordinates coincide in this limit! These equations can be combined as

$$J_C^\mu = e^{(K_x)^\mu{}_\nu \phi} e^{(K_y)^\nu{}_\rho \phi} J_A^\rho, \quad J_E^\mu = e^{(K_y)^\mu{}_\nu \phi} e^{(K_x)^\nu{}_\rho \phi} J_A^\rho \tag{35}$$

and finally

$$J_E^\mu = e^{(K_y)^\mu{}_\nu \phi} e^{(K_x)^\nu{}_\rho \phi} e^{-(K_y)^\rho{}_\sigma \phi} e^{-(K_x)^\sigma{}_\lambda \phi} J_C^\lambda \tag{36}$$

We are told that we can use the approximation

$$e^{\phi A} e^{\phi B} \simeq e^{\phi B} e^{\phi A} e^{\phi^2 [A, B]} \tag{37}$$

meaning

$$e^{-\phi K_y} e^{-\phi K_x} \simeq e^{-\phi K_x} e^{-\phi K_y} e^{\phi^2 [K_y, K_x]} \tag{38}$$

hence

$$J_E^\mu = e^{\phi^2 [K_y, K_x]^\mu{}_\nu} J_C^\nu \tag{39}$$

which we can rewrite using (33) as

$$J_E^\mu = e^{-\phi^2 (R_z)^\mu{}_\nu} J_C^\nu \tag{40}$$

We see that the boosts do not commute! In the leading order, their difference gives a rotation: Charlie sees the current rotated with respect to Eva! Explicitly,

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_E = \exp \left[-\phi^2 \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right] \begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_C \tag{41}$$

when we look at the 3-vector part. We can now expand the exponential:

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_E = \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + (-\phi^2) \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{(-\phi^2)^2}{2} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{(-\phi^2)^3}{3!} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \dots \right] \begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_C \tag{42}$$

hence we get

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_E = \begin{pmatrix} \cos(\phi^2) & -\sin(\phi^2) & 0 \\ \sin(\phi^2) & \cos(\phi^2) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_C \tag{43}$$

Name:	
Student ID:	



We have $\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_C = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ for part a and $\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_C = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ for part b, hence the answers are $(J_y)_E = \cos(\phi^2)$ and $(J_y)_E = 0$ respectively.

Bonus Question: 7: Understanding Special Relativity (30 points)

(EDWARDS TRANSFORMATIONS)

Although it is usually stated that the speed of light in any direction is same for any inertial observer in special relativity, this statement is not entirely true: what is same for any inertial observer is *the two-way speed of light*. It is entirely possible for physics to be unaltered even if the speed of light is $2c$ along some direction $\hat{n}$, and $2c/3$ along $-\hat{n}$, since the two-way speed of light in this case is still $\bar{v} = 2 \frac{v_1 v_2}{v_1 + v_2} = 2 \frac{(2c)(2c/3)}{(2c) + (2c/3)} = c$. Indeed, such a description (also known as non-standard synchronization) can be related to our standard Einstein's description via what-is-called the *Edwards transformation*.

If we were to write down the Lorentz force law in this particular non-standard synchronization, what would be different? Write down the *exact* mathematical form of the equation!

Solution 7.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

The functional form of an equation written in a Lorentz-covariant form would still be the same, as the underlying physics is still the same:

$$\frac{dP^\mu}{d\tau} = qF^{\mu\nu} U_\nu \quad (44)$$

This is the power of four-vector notation: what is written is directly related to what is physical.

In three vector notation, we would have a different equation now, as decomposing $F^{\mu\nu}$ into $\mathbf{E}$ and $\mathbf{B}$ becomes rather non-trivial:

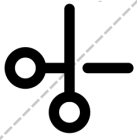
$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B} + \text{a term that involves } \hat{n}) \quad (45)$$

We can try to derive it, but what is the point? You should express laws in 4-vector notation, go to frames where computations are easiest, and then convert to 3-vector notation in that particular frame.

For any interested student who wants a challenge to derive 3-vector form in this particular synchronization, here is how you can start: choose your coordinate system such that $\hat{n}$ lies along x -axis. You take that in these (t, x, y, z) coordinates we use the standard Einstein's synchronization and Lorentz law is the traditional one:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (46)$$

Now, we would like to go to a new coordinate system in which speed of light is $2c$ in positive x direction



Name:	
Student ID:	

and $2c/3$ in negative x direction: call this new coordinate system (t', x', y', z') .

How should coordinates transform? It is not obvious but we are actually trying to do a shearing, i.e. an $SL(2, \mathbb{R})$ transformation that is not $SO(1, 1)$ (we are not preserving the metric structure but we should preserve volumes). Honestly, it is hard to understand any of these without a proper group theory background, so you might have a better chance with trial and error! At the end, you may get convinced that the transformation

$$\begin{pmatrix} t' \\ x' \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2c} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \end{pmatrix} \quad (47)$$

gives you

$$dx' = dx, \quad dt' = dt - \frac{1}{2c} dx = \left(1 - \frac{v}{2c}\right) dt \quad \text{for } v := \frac{dx}{dt} \quad (48)$$

with which we get

$$v' = \frac{dx'}{dt'} = \frac{dx}{dt'} = \frac{dx}{(1 - \frac{v}{2c})dt} = \frac{2cv}{2c - v} \quad (49)$$

This means that the transformation takes $v = c$ into $v' = 2c$ and $v = -c$ into $v' = -2c/3$: so the primed coordinates are precisely the coordinate system we were defining earlier! All we need to understand now is how the coordinate transformation in (47) changed $\mathbf{E}$ and $\mathbf{B}$ (note that (47) is not a Lorentz transformation!). Although this is not a long computation, there are a lot of subtleties so I won't go into it here: if you are interested, research the rest during your holiday as a private summer project (you may wanna check out the nice yet rather advanced discussion in [arXiv:physics/0510260](https://arxiv.org/abs/physics/0510260) as well)!

There is a really nice video about this by Veritasium; I highly suggest it: <https://youtu.be/pTn6Ewhb27k>. However, this video presents the situation as our lack of methods to define one-way speed of light; mathematically speaking, the situation is actually more general: whatever coordinate system you choose to work with, you can always make coordinate transformations in a way to change some things (such as one-way speed of light) while preserving other things (such as two-way speed of light and consistency with the principle of relativity); in fact, physics should be independent of how coordinate systems are chosen to begin with! This leads to what-is-called the diffeomorphism invariance, an essence of the General Relativity!

PS: You may find it interesting to note that the level of mathematics you already know is actually sufficient to derive (47). As we have explicitly discussed in class, Lorentz transformations have to be linear to respect principle of relativity, hence for some yet-to-be-determined coefficients a, b, c, d ,

$$v'(v) = \frac{dx'}{dt'} = \frac{adx + bdt}{cdx + ddt} = \frac{av + b}{cv + d} \quad (50)$$

is the most general transformation for the velocity along x direction. The velocity along $-x$ direction will then be

$$v'(-v) = \frac{-av + b}{-cv + d} \quad (51)$$

Name:	
Student ID:	



We want the subset of these transformations that preserve two-way-speed-of-light: this is the second postulate of special relativity. Hence, we need

$$\begin{aligned} \overline{v'} &= \overline{v} \\ 2 \frac{|v'(v)||v'(-v)|}{|v'(v)| + |v'(-v)|} &= 2 \frac{|v||-v|}{|v| + |-v|} \\ 2 \frac{\frac{av+b}{cv+d} \left| \frac{-av+b}{-cv+d} \right|}{\frac{-av+b}{-cv+d} + \left| \frac{-av+b}{-cv+d} \right|} &= v \end{aligned} \quad (52)$$

We can now analyze $\frac{-av+b}{-cv+d} > 0$ and $\frac{-av+b}{-cv+d} < 0$ cases separately; in fact, there is no solution for the first case. A straightforward algebra for the second case, i.e. $\left| \frac{-av+b}{-cv+d} \right| = -\frac{-av+b}{-cv+d}$, yields

$$b = 0, \quad a = d, \quad c \in \mathbb{R} \quad (53)$$

meaning that for arbitrary α, β ,

$$\begin{pmatrix} t' \\ x' \end{pmatrix} = \alpha \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \end{pmatrix} \quad (54)$$

is the most general Edwards transformation that changes velocities along x direction, of which (47) is a special case.

« « « Congratulations, you have made it to the end! » » »